

PRODUCT OVERVIEW — CONFIDENTIAL

KayCare HMS

Enterprise Hospital Management System

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SECTION 01

Executive Summary

KayCare HMS is a comprehensive, enterprise-grade Hospital Management System designed to bring every dimension of healthcare operations onto a single, unified digital platform. From patient registration and clinical consultations through to laboratory diagnostics, pharmacy dispensing, inpatient nursing care, radiology, and financial billing — KayCare HMS handles it all in one place.

The system is built for healthcare facilities that require not just a tool, but a dependable operational backbone: one that maintains clinical data accuracy, enforces financial transparency, and keeps every department in sync with one another in real time.

CORE VALUE PROPOSITION

KayCare HMS eliminates the fragmentation that plagues traditional healthcare IT environments, where separate systems for records, lab, pharmacy, and billing operate independently and create gaps in care and revenue. In its place, it delivers a tightly integrated clinical and financial engine with role-based access, AI-assisted workflows, and full audit traceability.

The platform is designed to accommodate facilities of varying sizes — from single-branch clinics to multi-facility hospital groups — through a multi-tenant architecture that keeps each organisation's data completely isolated and secure, while running on shared infrastructure for cost efficiency.

AT A GLANCE

10+ integrated clinical and administrative modules — covering the full patient journey from first registration to final billing — with built-in AI assistance, standards-based laboratory connectivity (HL7), and a complete role-based access control system governing more than 9 distinct user types.

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SECTION 02

The Problem We Solve

Many healthcare facilities today operate with a patchwork of disconnected software products — one for patient records, another for laboratory results, a separate billing platform, and often manual processes in between. This fragmentation creates compounding operational problems that directly affect clinical quality, financial performance, and staff efficiency.

Challenge	The Fragmented Approach	The KayCare HMS Solution
Data Accuracy	Lab results and medication orders are manually transferred between systems, introducing errors.	Orders placed by a physician flow instantly to the lab worklist and pharmacy queue — no re-entry required.
Revenue Leakage	Tests and medications frequently escape billing because charge capture is disconnected from clinical activity.	Every clinical event — from a dispensed tablet to a diagnostic test — is automatically captured on the patient's bill.
Pharmacy Stock	Dispensing updates require separate stock entries, causing inventory discrepancies.	Dispensing a prescription instantly deducts the exact quantity from live inventory balances.
Financial Visibility	Accounts receivable and revenue performance data require manual reconciliation across systems.	A live revenue dashboard, AR aging reports, and insurance claim tracking provide real-time financial clarity.

Challenge	The Fragmented Approach	The KayCare HMS Solution
Clinical Documentation	Handwritten or unstructured notes are inaccessible across departments and carry compliance risks.	Structured SOAP documentation, ICD-10 coding, allergy alerts, and discharge summaries are all digitised and searchable.

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SECTION 03

Core Modules & Capabilities

KayCare HMS is organised into interconnected modules, each managing a distinct domain of healthcare operations. Together they form a seamless clinical and administrative workflow.

Patient EMR & Outpatient Consultations

The Electronic Medical Record serves as the central repository for every patient's clinical history. Each patient is assigned a unique Medical Record Number (MRN) upon registration, and their complete clinical timeline — from demographics and allergies through to consultations, prescriptions, lab results, and vitals — is available in one consolidated profile.

- Comprehensive patient registration with demographic profiles and emergency contacts
- Longitudinal vital sign tracking (Blood Pressure, Heart Rate, Temperature, O2 Saturation, BMI, Respiratory Rate) with trend visualisation
- Allergy recording with automated prescription alerts to prevent unsafe drug dispensing
- Structured SOAP consultation documentation (Subjective, Objective, Assessment, Plan)
- Integrated ICD-10 diagnosis coding with primary and secondary code assignment
- Patient referral management to internal departments or external specialists
- Secure document storage for uploaded clinical files and reports

Inpatient Department (IPD) & Ward Management

The inpatient module manages the full lifecycle of a hospital admission — from bed assignment on arrival through to discharge summary generation. Ward configurations, bed availability, and admission status are visible in real time across clinical staff.

- Dynamic ward and bed management across General, Semi-Private, Private, ICU, and HDU categories
- Admission workflows including attending physician assignment and ward transfers
- Automated daily room rate charge capture linked directly to the patient bill
- Formal discharge summary generation

Nursing Workspace

Nursing staff have a dedicated workspace for shift-based clinical documentation and medication administration, keeping care teams informed and accountable across handovers.

- Medication Administration Record (MAR) with verification of dose timing, administrator, and status (Administered, Held, Refused)
- Nursing shift notes for structured clinical observations
- Vital sign monitoring sheets linked to patient admission records

SECTION 03 — CONTINUED

Core Modules & Capabilities

Laboratory Information System (LIS)

The laboratory module digitises the entire diagnostic workflow, from the moment a physician places an order through to a validated, signed-off result. It supports both manual result entry and direct machine connectivity for facilities with automated laboratory analysers.

- Physician-driven lab order placement linked to the patient record and billing ledger
- Sample accessioning with tracked lifecycle: Ordered → Sample Received → Resulted → Signed Off
- HL7 MLLP connectivity — enabling automated result ingestion directly from laboratory analysers, eliminating manual transcription
- Critical result flagging with mandatory call-log documentation to confirm physician notification
- Configurable lab test catalogue with reference ranges and unit definitions
- AI-assisted lab result interpretation (see Section 04)

Radiology & Diagnostic Imaging

Radiology orders can be placed directly from within the system and routed to a dedicated radiologist or technician worklist, ensuring imaging studies are tracked and results are captured without delays.

- Radiology order placement for X-Ray, Ultrasound, CT, MRI, and configurable imaging modalities
- Radiology technician worklist and order lifecycle management
- Integrated DICOM image viewing — diagnostic images can be reviewed directly within the browser without additional software
- Configurable imaging procedure catalogue

Pharmacy, Drug Formulary & Inventory

The pharmacy module connects clinical prescribing directly to dispensing and inventory management, providing end-to-end visibility over pharmaceutical stock and ensuring that regulatory obligations around controlled substances are met.

- Electronic prescribing from consultation records with dosage validation and refill controls
- Prescription dispensing with real-time, automated inventory deductions
- Controlled Substances Register (CS Register) with dedicated, audited transaction logs for schedule drugs
- Drug formulary management with payer-specific pricing support
- Supplier catalogue and purchase order management with Goods Received Note (GRN) logging
- Automated low-stock reorder alerts
- AI-powered drug interaction and safety screening (see Section 04)

Billing, Revenue Cycle & Insurance Claims

Every clinical event in KayCare HMS feeds into a centralised billing ledger automatically. Billing staff work from a complete, itemised patient invoice that has been populated in real time as care was delivered, minimising revenue leakage and reducing manual billing effort.

- Centralised charge capture aggregating consultation fees, lab tests, medications, ward bed rates, and nursing services
- Line-item adjustments, authorised discounts, and partial payment support
- Credit notes and refund processing with full audit trails
- Insurance claims generation and lifecycle tracking
- Payer management with configurable tariff schedules
- Accounts Receivable (AR) Aging reports (0–30, 31–60, 61–90, 90+ day buckets)
- Revenue dashboard with financial performance metrics

SECTION 04

AI-Powered Clinical Assistance

KayCare HMS incorporates a suite of AI-assisted clinical tools powered by Google's Gemini language model. These tools are designed to augment the work of clinical and pharmacy staff — reducing administrative burden, surfacing insights faster, and improving the quality of patient communication — while keeping the clinician firmly in control of all final decisions.

SOAP Note Copilot

Physicians can enter raw, unstructured consultation notes in natural language. The AI automatically organises them into a structured SOAP format (Subjective, Objective, Assessment, Plan) and suggests appropriate ICD-10 diagnosis codes — dramatically reducing documentation time.

Patient Summary Generator

After a consultation is completed, the AI generates a plain-English take-home summary for the patient — translating clinical SOAP notes into clear instructions covering their diagnosis, medications, self-care steps, and warning signs to watch for.

Lab Result Interpreter

Requested on any completed lab panel, this tool provides the ordering physician with a structured clinical interpretation — flagging abnormal values, explaining potential pathophysiological correlations, and recommending next investigative steps.

Drug Safety & Interaction Screener

Before a prescription is dispensed, the AI reviews the full medication cart for drug-drug interactions, controlled substance flags, and generates patient counselling guidance — supporting safe dispensing practice at the point of care.

IMPORTANT NOTE

All AI-generated outputs within KayCare HMS are clearly identified as AI-assisted suggestions. Final clinical decisions remain the full responsibility of the treating clinician or licensed pharmacist. The system does not act autonomously on any clinical recommendation.

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SECTION 05

User Roles & Access Control

Access to every area of KayCare HMS is governed by a granular, role-based access control (RBAC) system. Each user is assigned a role that determines precisely which modules and actions they can access — ensuring that sensitive clinical and financial data is only visible to staff with a legitimate operational need.

Role	Scope of Access
Super Admin	Platform-level administration — manages all tenants, facility-wide settings, and platform audit logs. Full system access.
Administrator	Facility-level administration — manages staff accounts, departments, service catalogues, ward setup, drug formulary, payer tariffs, and has read access across all modules.
Doctor	Consultations, SOAP documentation, prescription writing, lab order placement, lab result review, radiology orders, inpatient admissions, referrals, and clinical reports.
Nurse	Vital sign recording, nursing notes, medication administration records, consultation support, inpatient wards, and patient documents.
Pharmacist	Prescription review and dispensing, drug inventory management, purchase orders, supplier management, CS Register, and reorder alerts.
Lab Technician	Lab order worklist management, sample processing, result entry, HL7 inbox for analyser connectivity, and lab result sign-off.
Billing Officer	Patient billing, invoice management, insurance claims, credit notes, refunds, AR aging, and the revenue dashboard.
Receptionist	Patient registration, appointment scheduling, inpatient admissions coordination, and basic billing operations.

Role assignments are managed by facility administrators and can be adjusted at any time. All login activity, access patterns, and data interactions are recorded in a tamper-evident audit log for compliance and accountability purposes.

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SECTION 06

Technology Stack

KayCare HMS is built on a carefully selected set of proven, enterprise-grade technologies. Each choice reflects a balance of performance, reliability, long-term vendor support, and suitability for the demands of healthcare data management.

FRONTEND**React 18 with TypeScript**

React is the most widely adopted framework for building interactive web applications, backed by Meta and a vast ecosystem. TypeScript adds compile-time type safety, which significantly reduces runtime errors in a clinical context where reliability is non-negotiable. The result is a fast, responsive single-page interface that behaves consistently across devices and browsers.

BACKEND API**ASP.NET Core 8**

Microsoft's ASP.NET Core is a mature, high-performance server framework with strong enterprise adoption in the healthcare sector. Version 8 delivers excellent throughput for concurrent API requests, native support for asynchronous processing, and a well-established security model — making it a sound choice for a platform handling sensitive patient data at scale.

DATABASE**PostgreSQL**

PostgreSQL is one of the world's most reliable open-source relational databases, trusted across the financial, healthcare, and public sector industries. It provides full ACID transaction guarantees — meaning data is never partially saved or corrupted — and handles complex concurrent queries with robust concurrency controls essential for multi-department hospital usage.

FILE & DOCUMENT STORAGE**Supabase Storage**

Supabase Storage provides secure, scalable object storage with fine-grained access control via signed URLs. It is used for DICOM medical images, patient documents, and generated PDFs. Files are never exposed publicly; every access is authorised through the application layer.

DATA ACCESS LAYER**Entity Framework Core 8**

Entity Framework Core is Microsoft's official object-relational mapper, providing a structured and type-safe interface between the application code and the database. Its Global Query Filter feature is central to KayCare's multi-tenancy model, automatically applying tenant isolation rules to every database query without requiring manual filtering in business logic.

AI INTEGRATION**Google Gemini API**

Google's Gemini model is integrated to power the platform's AI clinical assistance features. It was selected for its strong performance on structured medical language tasks — SOAP formatting, ICD-10 code suggestion, lab interpretation — as well as its reliable API infrastructure and low-latency response times suitable for real-time clinical workflows.

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SECTION 07

Security, Compliance & Data Integrity

Healthcare data requires the highest standards of security architecture and regulatory compliance. KayCare HMS is built with these requirements as foundational design priorities, not afterthoughts.

Authentication & Account Security

- JWT-based stateless authentication with configurable token expiry
- Enforced password change on first login for all new accounts
- Account lockout after repeated failed login attempts, with configurable thresholds
- Role-based route protection — unauthenticated and unauthorised access attempts are rejected at both the API and UI layer

HIPAA-Aligned Audit Logging

Every meaningful action taken within the system is recorded in an immutable audit log. This log captures the staff member involved, the patient record accessed, the nature of the action (creation, modification, deletion, or view), the exact API endpoint, the client IP address, and a precise timestamp. For financial and medical record changes, the audit log stores a full before-and-after snapshot of the data — ensuring complete historical accountability and supporting regulatory audit reviews.

Financial Transaction Integrity

All billing checkouts, insurance claim processing, credit note issuance, and inventory stock deductions are executed within explicit database transactions. If any step in a multi-part financial operation encounters an error, the entire operation rolls back cleanly — eliminating partial saves, billing discrepancies, and inventory mismatches.

CONCURRENCY CONTROL

KayCare HMS implements concurrency protection mechanisms to prevent race conditions in scenarios where multiple staff members may act on the same resource simultaneously — for example, two pharmacists attempting to dispense the same stock, or two nurses booking the same bed. The system resolves these conflicts automatically without data loss.

Data Isolation

For multi-facility deployments, each organisation's data is categorically isolated at the database query level. It is architecturally impossible for one facility's staff to view or access another facility's patient data, financial records, or operational information — even when both run on the same underlying infrastructure.

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SECTION 08

Multi-Tenancy & Scalability

KayCare HMS is architected as a true multi-tenant platform, meaning a single deployment can serve multiple independent healthcare organisations simultaneously — each operating with complete data separation and their own configuration, staff, and branding.

This architecture delivers significant advantages for both healthcare groups managing multiple facilities and for the platform operator:

- Each tenant (facility) operates in a fully isolated data environment, even though the underlying infrastructure is shared
- Tenant isolation is enforced at the database access layer — not just in the application logic — making cross-tenant data exposure architecturally prevented
- New facilities can be onboarded by a Super Admin without any code deployment or infrastructure changes
- Each facility can independently configure its service catalogue, payer tariffs, drug formulary, ward layout, and user roles
- Platform-level reporting and audit logs give operators visibility across all tenants without exposing individual facility data to one another

ARCHITECTURE PATTERN

KayCare HMS uses a Modular Monolith architecture — combining the clean internal separation of microservices with the operational simplicity of a single deployable unit. This means the system is straightforward to host, maintain, and scale, while remaining highly adaptable to changing functional requirements over time.

SECTION 09

Data Migration

Transitioning from an existing system to KayCare HMS is executed through a structured, three-phase migration protocol designed to preserve data integrity and minimise any disruption to clinical operations. The process is designed so that no historical data — patient records, financial balances, or inventory — is lost in the transition.

1

Secure Extraction & Audit

Legacy database exports covering patient demographics, consultation history, item catalogues, and outstanding financial balances are extracted into an encrypted staging environment. All extraction runs during off-peak hours so clinical operations are completely unaffected.

2

Schema Mapping, Scrubbing & Normalisation

Legacy records are systematically cleaned, deduplicated, and transformed to align with KayCare HMS's relational data models. Automated integrity checks verify that all clinical notes reference valid patient records and that financial figures reconcile before any data is loaded into the production environment.

3

Parallel Validation & Cutover Sign-Off

A validation layer compares historical totals and ledger balances between the legacy system and KayCare HMS. Clinical and administrative staff review the migrated data during a parallel-run period. Only upon satisfactory sign-off is production access transferred, ensuring clinical staff have complete patient history available from Day One.

This approach ensures zero data loss, zero clinical downtime, and a verifiable audit trail of the entire migration process.

SECTION 10

Deployment & Hosting

KayCare HMS is designed for cloud-native deployment, offering flexibility in hosting environment to match the infrastructure preferences and compliance requirements of the organisation.

Backend Deployment

The server-side API is packaged as a Docker container image, making it portable and host-agnostic. It can be deployed on any container-capable cloud platform, including AWS Elastic Container Service, Microsoft Azure App Service, Google Cloud Run, or managed container hosting platforms such as Render or Railway. The containerised approach ensures consistent, reproducible behaviour across environments.

Frontend Delivery

The web-based client application is a compiled static build, optimised for delivery via global Content Delivery Network (CDN) infrastructure. This ensures fast load times regardless of where end users are located. Compatible CDN hosting platforms include Vercel, Netlify, and Cloudflare Pages.

Database Hosting

PostgreSQL can be hosted on any managed database platform — including AWS RDS, Azure Database for PostgreSQL, Supabase, or Render's managed Postgres service. Automated daily backups, point-in-time recovery, and connection pooling are recommended configuration standards for production environments.

ON-PREMISES DEPLOYMENT

For facilities with on-premises hosting requirements — whether due to regulatory constraints, network architecture, or data residency policies — KayCare HMS can equally be deployed within a private server environment. The containerised backend and static frontend are infrastructure-agnostic and require no external internet connectivity to function once deployed.

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